# SMARTX: Porewater SO4/Cl

September 2024 Samples 333-463

2025-10-03

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##Add Required Packages

## 0.1 Run Information

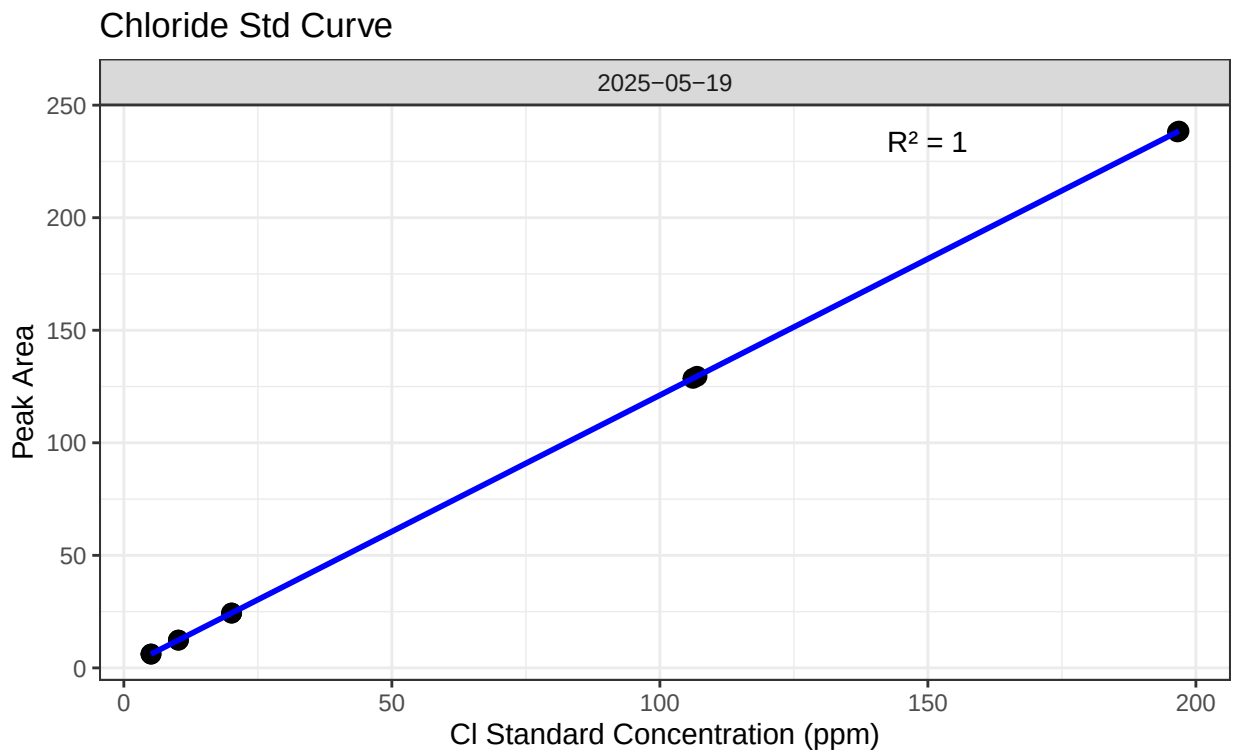
```
###### Run information - PLEASE CHANGE
  Date_Run = "2025-05-19"  #Date that instrument was run
  Run_by = "Zoe Read"  #Instrument user
  Script_run_by = "Zoe Read" #Code user
  sample_year <- "2024"
  sample_month <- "Sept"
  Site = "GCReW"
  run_notes = " " #any notes from the run
  samples <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names
  samples_pattern <- paste(samples, collapse = "|")
    #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
  raw_file_name_cl = "Raw Data/SMARTX_202409_333-463_Cl.txt"
  raw_file_name_so4 = "Raw Data/SMARTX_202409_333-463_SO4.txt"

#file path and name of processed data file
  processed_file_name = "Processed Data/GCReW_SMARTX_PW_Processed_Cl_SO4_202409_333-463.csv"

###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = here("..", "GCReW", "GCReW_Project_Treatment_Metadata.csv")

  #qaqc log file path for this year
  Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2024.csv"
```

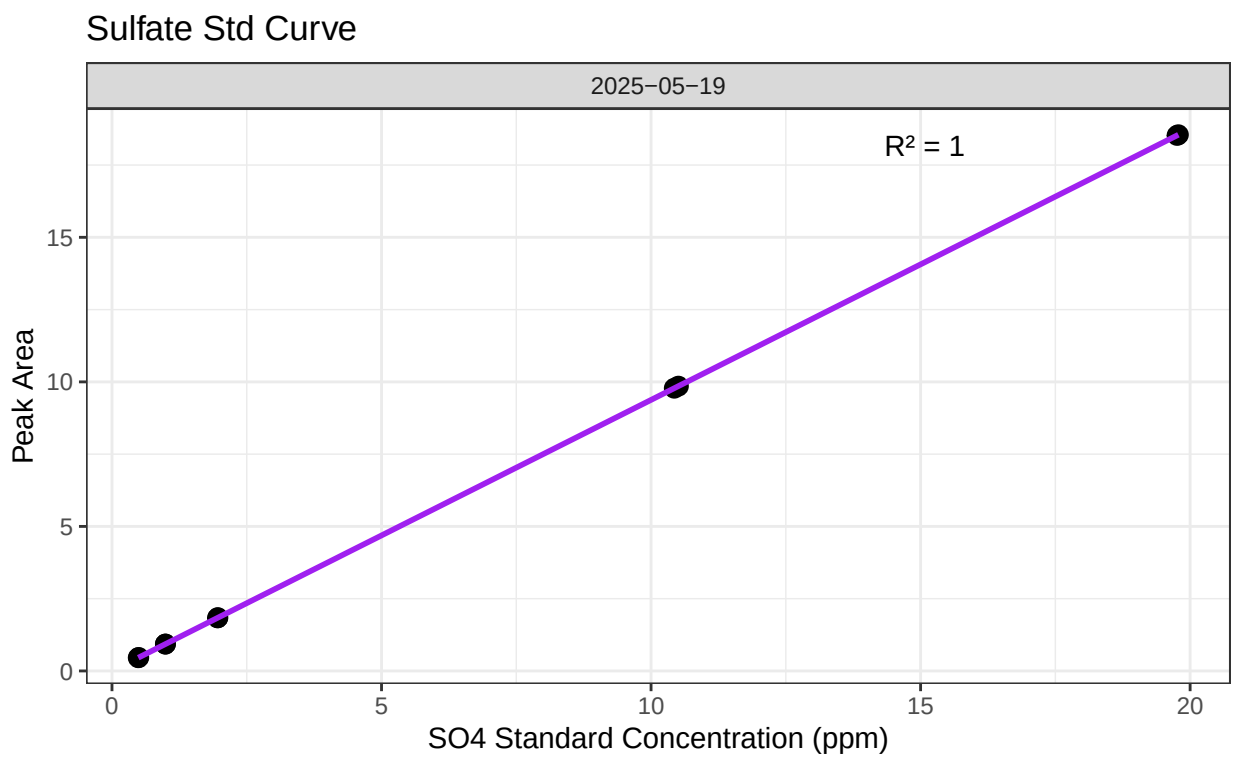
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

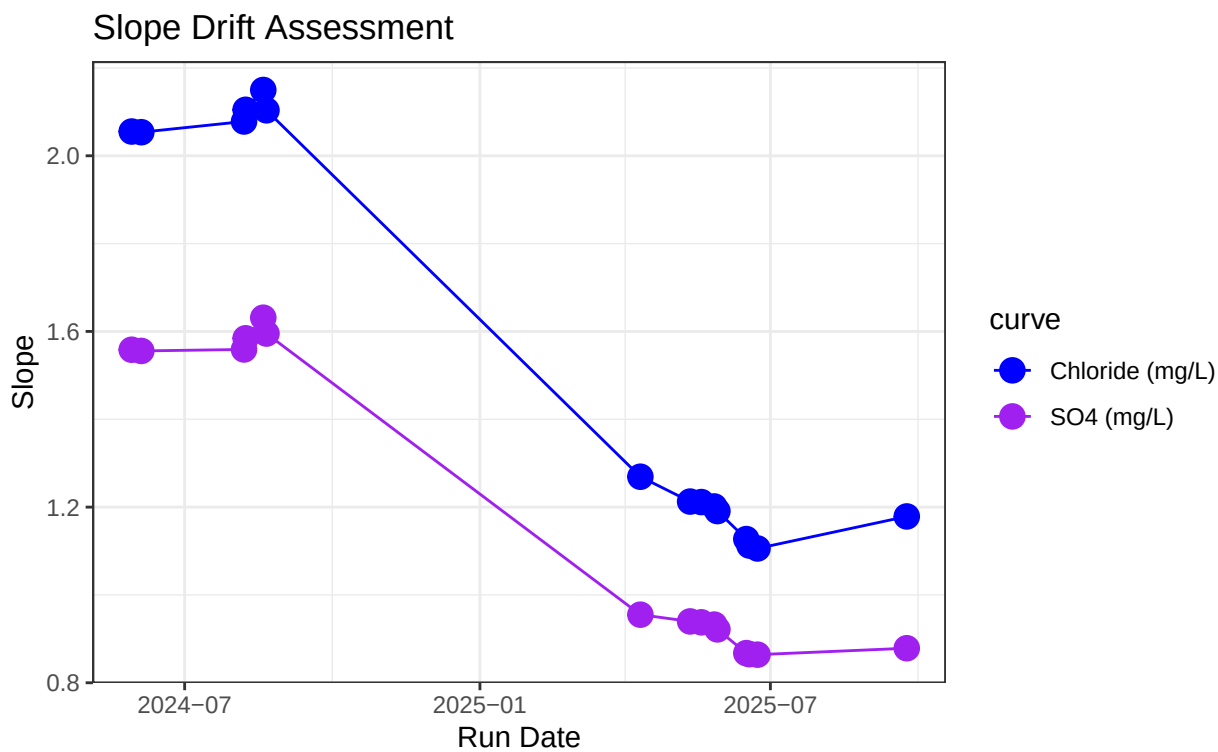
##Import Sample Data

## 0.2 Assess Standard Curves



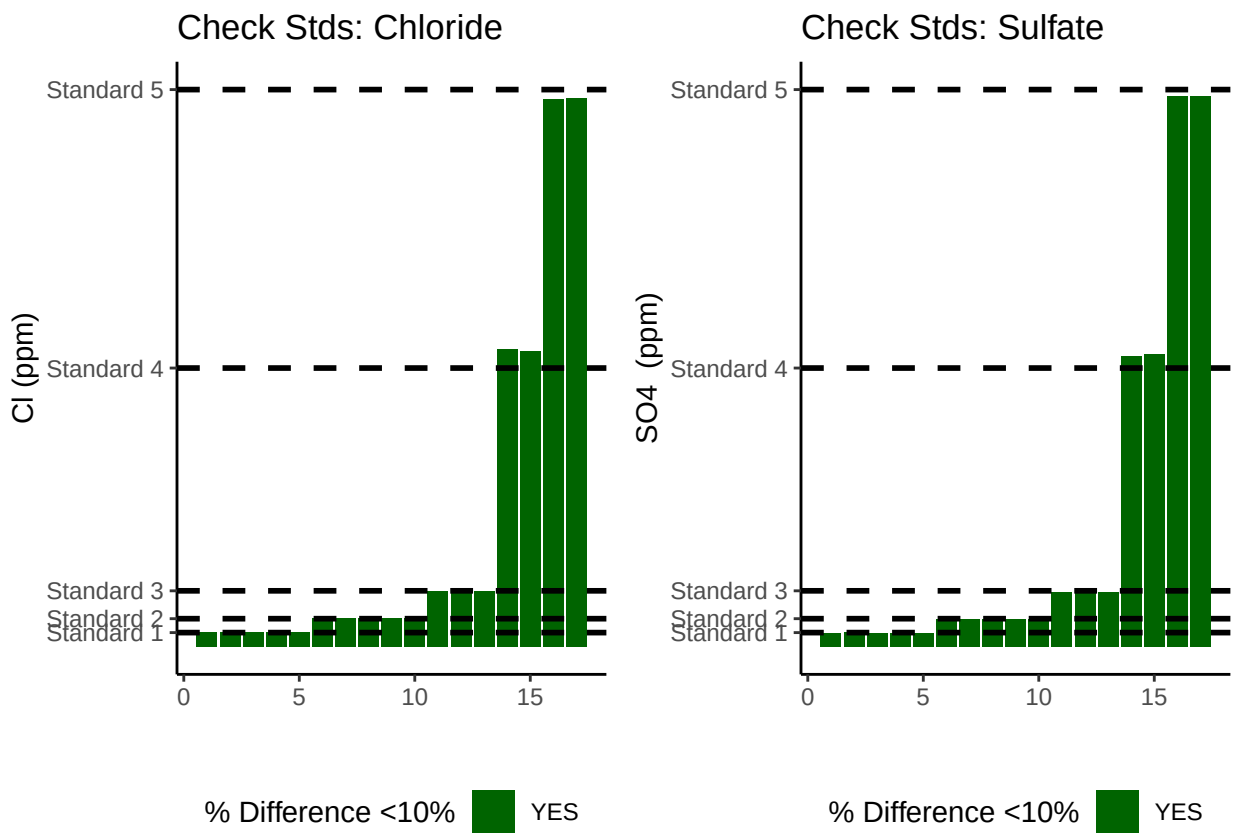

```
## [1] "QAQC log file exists and has been read into the code."
```




```
## [1] "Cl Curve r2 GOOD"

## [1] "SO4 Curve r2 GOOD"
```

## 0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl  sd_Cl   cv_Cl flag_Cl
##   <chr>        <dbl>  <dbl>   <dbl> <chr>
## 1 Standard 1    5.07 0.0147 0.00290 Chloride Check Standard RSD within Range - ~
## 2 Standard 2   10.2  0.0311 0.00307 Chloride Check Standard RSD within Range - ~
## 3 Standard 3   20.1  0.0426 0.00212 Chloride Check Standard RSD within Range - ~
## 4 Standard 4  107.   0.529  0.00496 Chloride Check Standard RSD within Range - ~
## 5 Standard 5  197.   0.202  0.00103 Chloride Check Standard RSD within Range - ~

## # A tibble: 5 x 5
##   sample_ID  mean_SO4  sd_SO4   cv_SO4 flag_SO4
##   <chr>         <dbl>   <dbl>    <dbl> <chr>
## 1 Standard 1    0.495 0.00589 0.0119   Sulfate Check Standard RSD within Range ~
## 2 Standard 2    0.992 0.00221 0.00223  Sulfate Check Standard RSD within Range ~
## 3 Standard 3    1.96  0.00560 0.00285  Sulfate Check Standard RSD within Range ~
## 4 Standard 4   10.5   0.0540  0.00515  Sulfate Check Standard RSD within Range ~
## 5 Standard 5   19.8   0.0174  0.000880 Sulfate Check Standard RSD within Range ~

## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"

## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



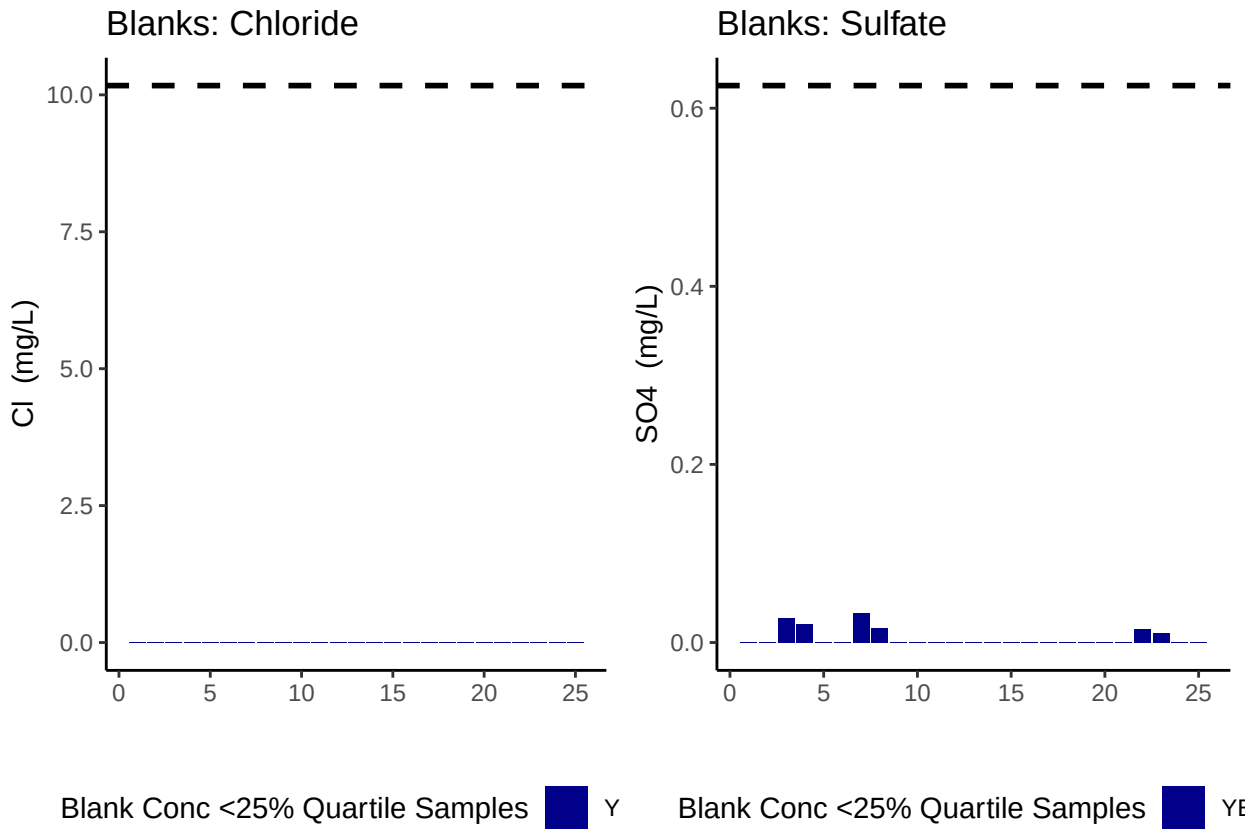

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

## 0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

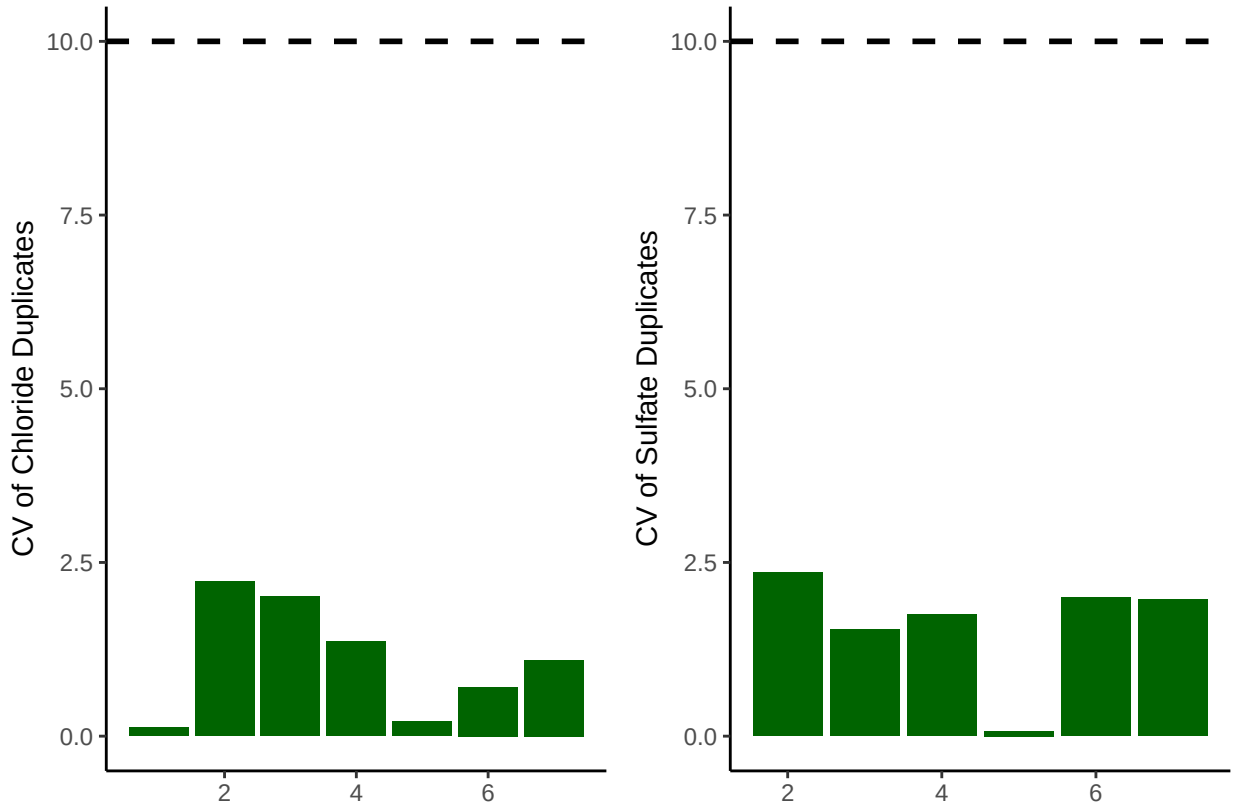
```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



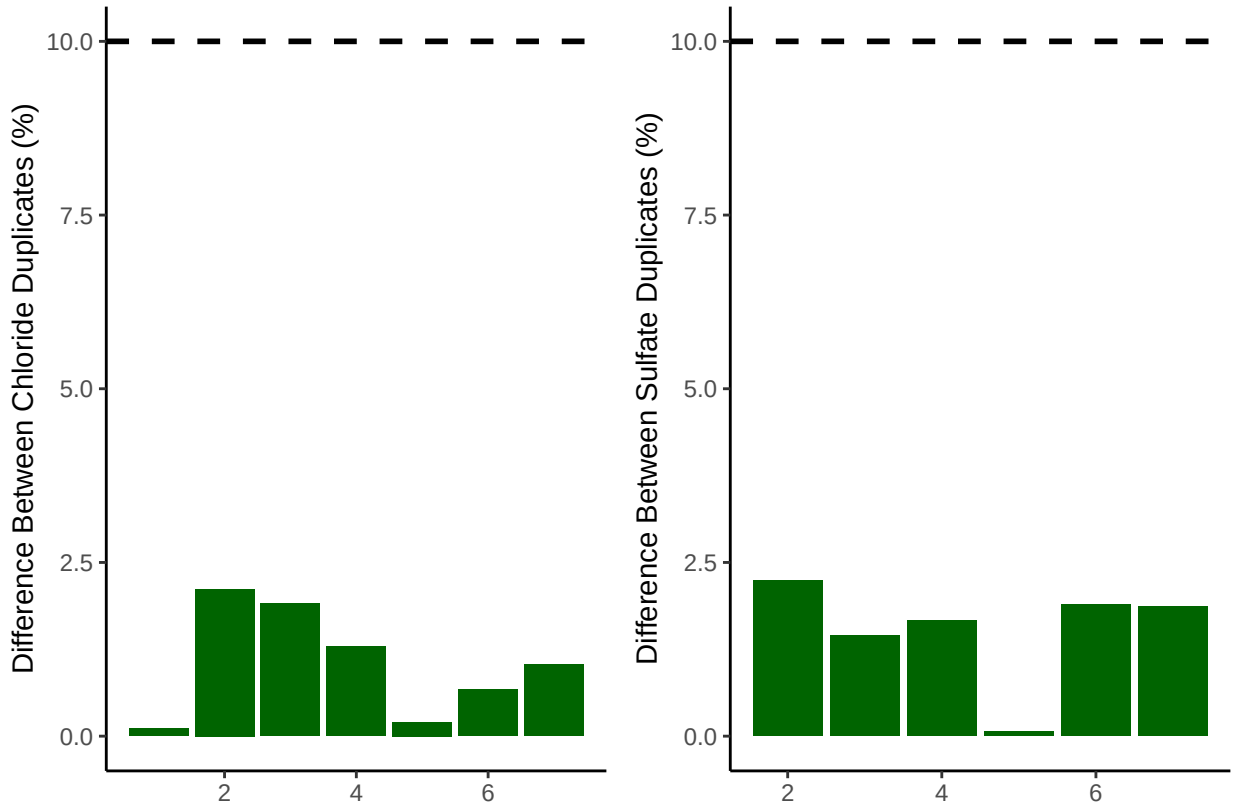

```
## Chloride blanks mean ppm:

## [1] 0.00114

## Sulfate blanks mean ppm:

## [1] 0.004944
```

## 0.5 Assess Duplicates

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"

## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```



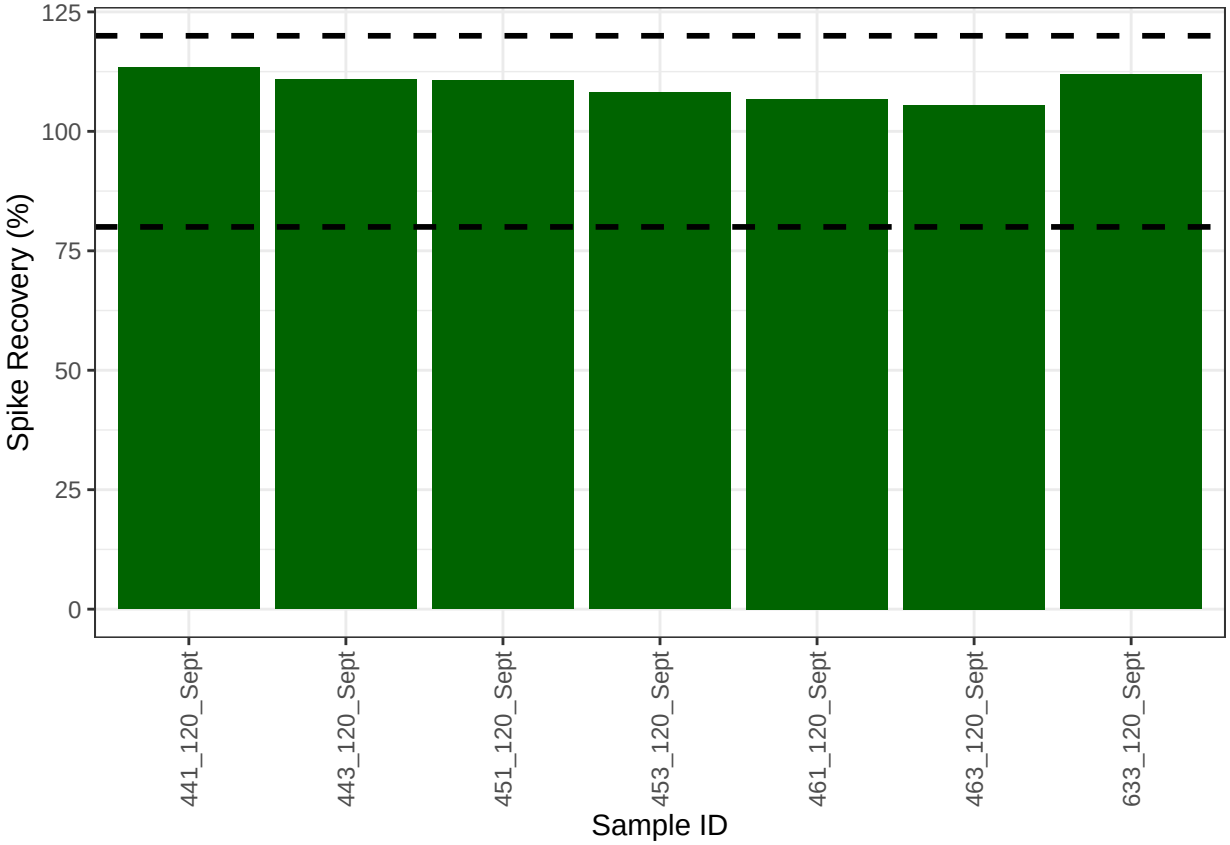

```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

## 0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

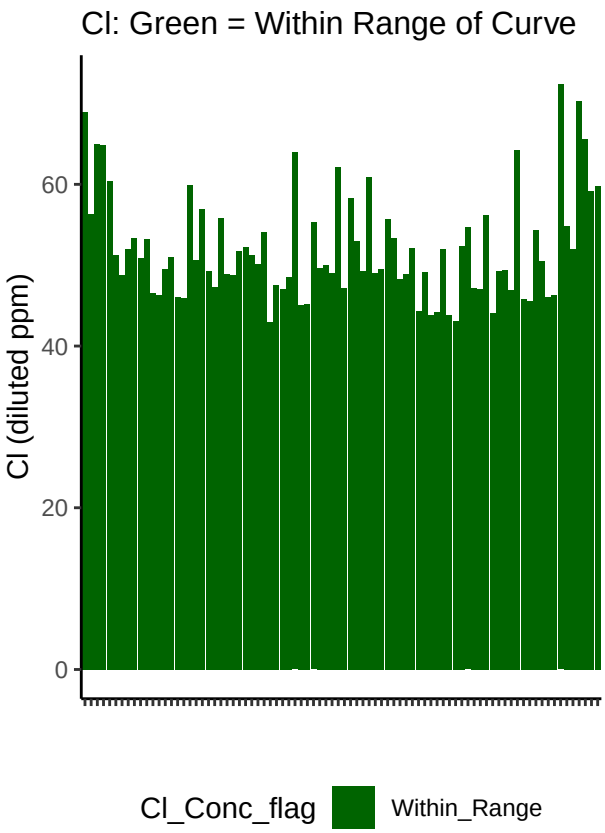
## 0.7 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

## 0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```



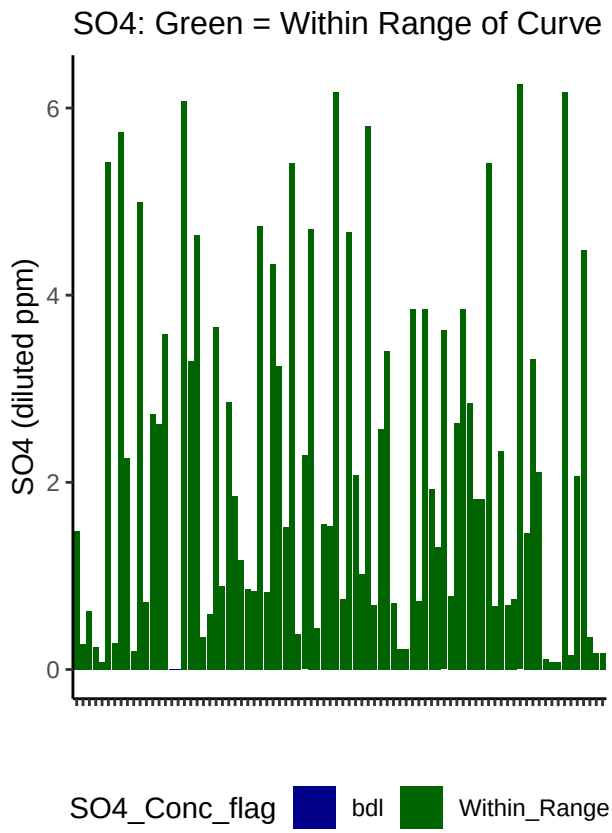




Table 1: SO4 samples

| SO4_Conc_flag | Percent_samples |
|---|---|
| Within_Range | 97.619048 |
| bdl | 2.380952 |

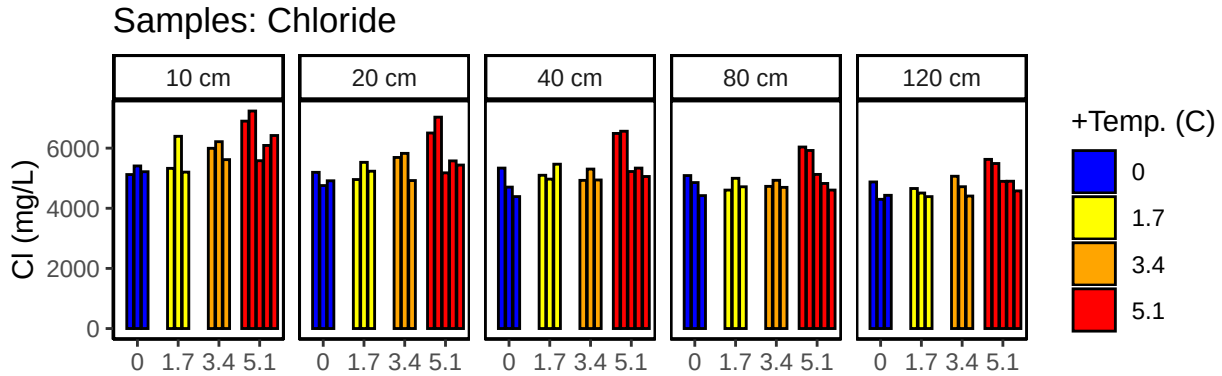
Table 2: Cl samples

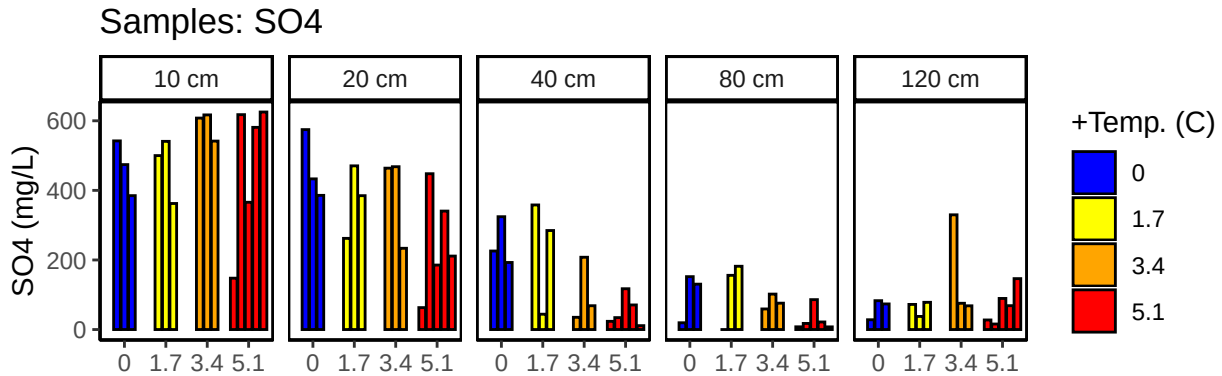
| Cl_Conc_flag | Percent_samples |
|---|---|
| Within_Range | 100 |

## 0.9 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```

## 0.10 Visualize Data by Plot




## 0.11 Export Processed Data

#end